J10

RJ45 shielded, no magnetics

CHASSIS >2	SHIELD	
	P1	A_L1
	P2	A_L2
	P3	A_L3
	P6	A_L6
	P4	A_L4
	P5	A_L5
	P7	A_L7
	P8	A_L8

T1

BARRIER

1000BASE-T PoE magnetics, 4 CT

A_M1	LINE_A_P	PHY_A	A_MDI_A >2
A_M2	LINE_A_N	PHY_B	A_MDI_B >2
A_M3	LINE_B_P	PHY_C	A_MDI_C >2
A_M6	LINE_B_N	PHY_D	A_MDI_D >2
A_M4	LINE_C_P	CT_A	CT_A >4
A_M5	LINE_C_N	CT_B	CT_B >4
A_M7	LINE_D_P	CT_C	CT_C >4
A_M8	LINE_D_N	CT_D	CT_D >4
CHASSIS >2	BOB_SMITH_HV_CAP		

K1

BARRIER

DPDT signal relay, pair 1-2

A_L1	COM_A	NC_A	A_M1
A_L2	COM_B	NC_B	A_M2
+5V >3,4,5,7	COIL_SUPPLY	NO_A	A_T1
		NO_B	A_T2
		COIL	RELAY_TEST_EN >6

PROT1

BARRIER

8-line protection network

A_T1	IN1	OUT1	WM1 >3
A_T2	IN2	OUT2	WM2 >3
A_T3	IN3	OUT3	WM3 >3
A_T6	IN6	OUT4	WM4 >3
A_T4	IN4	OUT5	WM5 >3
A_T5	IN5	OUT6	WM6 >3
A_T7	IN7	OUT7	WM7 >3
A_T8	IN8	OUT8	WM8 >3
GND >2,3,4,5,6,7	CLAMP_RTN		

K2

BARRIER

DPDT signal relay, pair 3-6

A_L3	COM_A	NC_A	A_M3
A_L6	COM_B	NC_B	A_M6
RELAY_TEST_EN >6	COIL	NO_A	A_T3
+5V >3,4,5,7	COIL_SUPPLY	NO_B	A_T6

K3

BARRIER

DPDT signal relay, pair 4-5

A_L4	COM_A	NC_A	A_M4
A_L5	COM_B	NC_B	A_M5
RELAY_TEST_EN >6	COIL	NO_A	A_T4
+5V >3,4,5,7	COIL_SUPPLY	NO_B	A_T5

K4

BARRIER

DPDT signal relay, pair 7-8

A_L7	COM_A	NC_A	A_M7
A_L8	COM_B	NC_B	A_M8
RELAY_TEST_EN >6	COIL	NO_A	A_T7
+5V >3,4,5,7	COIL_SUPPLY	NO_B	A_T8

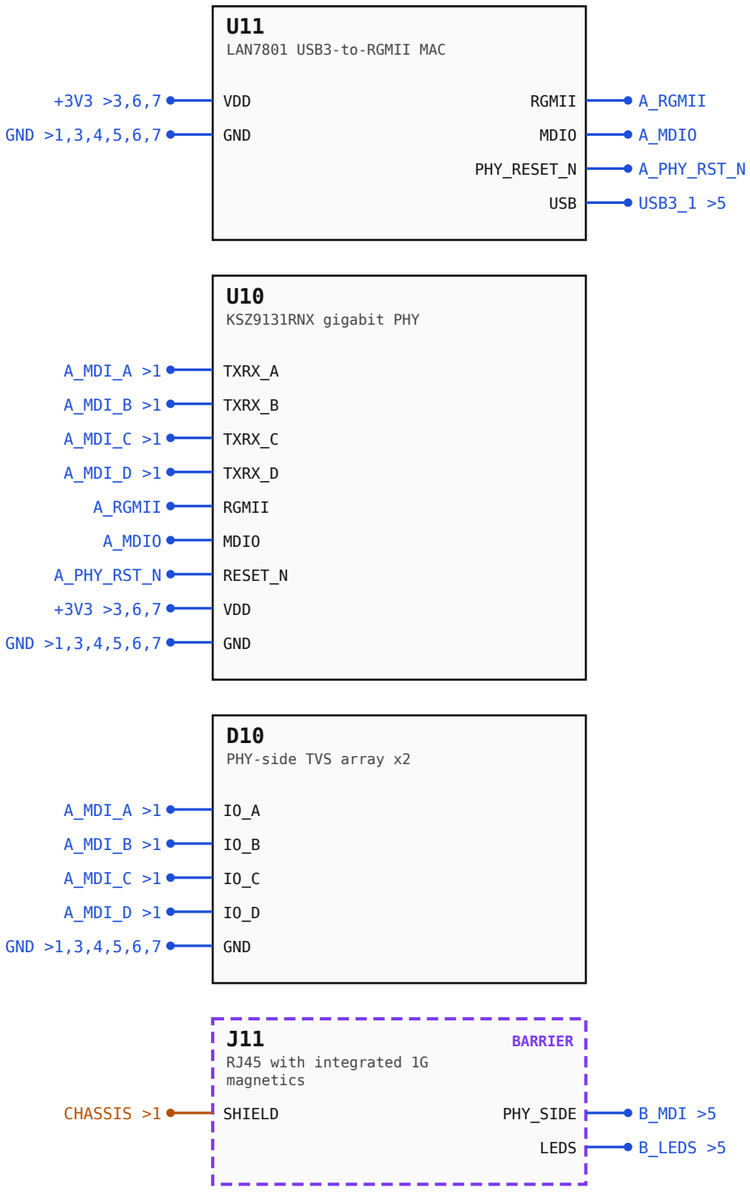
NetScope X1 main board - block schematic, rev A draft

Port A: jack, relays, PoE magnetics, protection

Pre-hardware. Not a KiCad schematic. Not ERC-checked. No board exists.

SYS system side · LINE cable conductors · POE network-powered

Block level: function-named pins, not package pins



NetScope X1 main board - block schematic, rev A draft

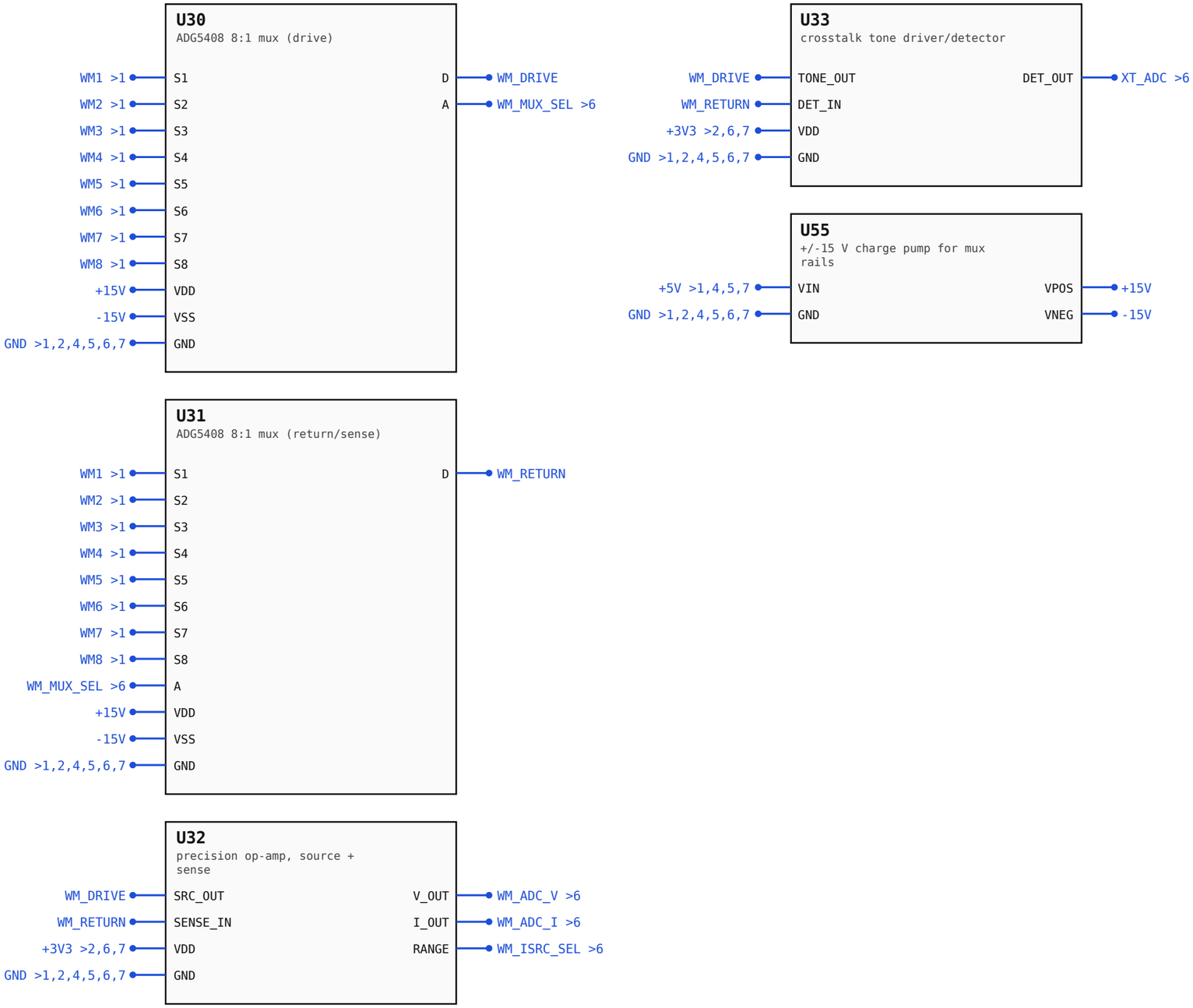
Test-port MAC and PHY, uplink port

Pre-hardware. Not a KiCad schematic. Not ERC-checked. No board exists.

SYS system side · LINE cable conductors · POE network-powered

Sheet 2 of 10

Block level: function-named pins, not package pins



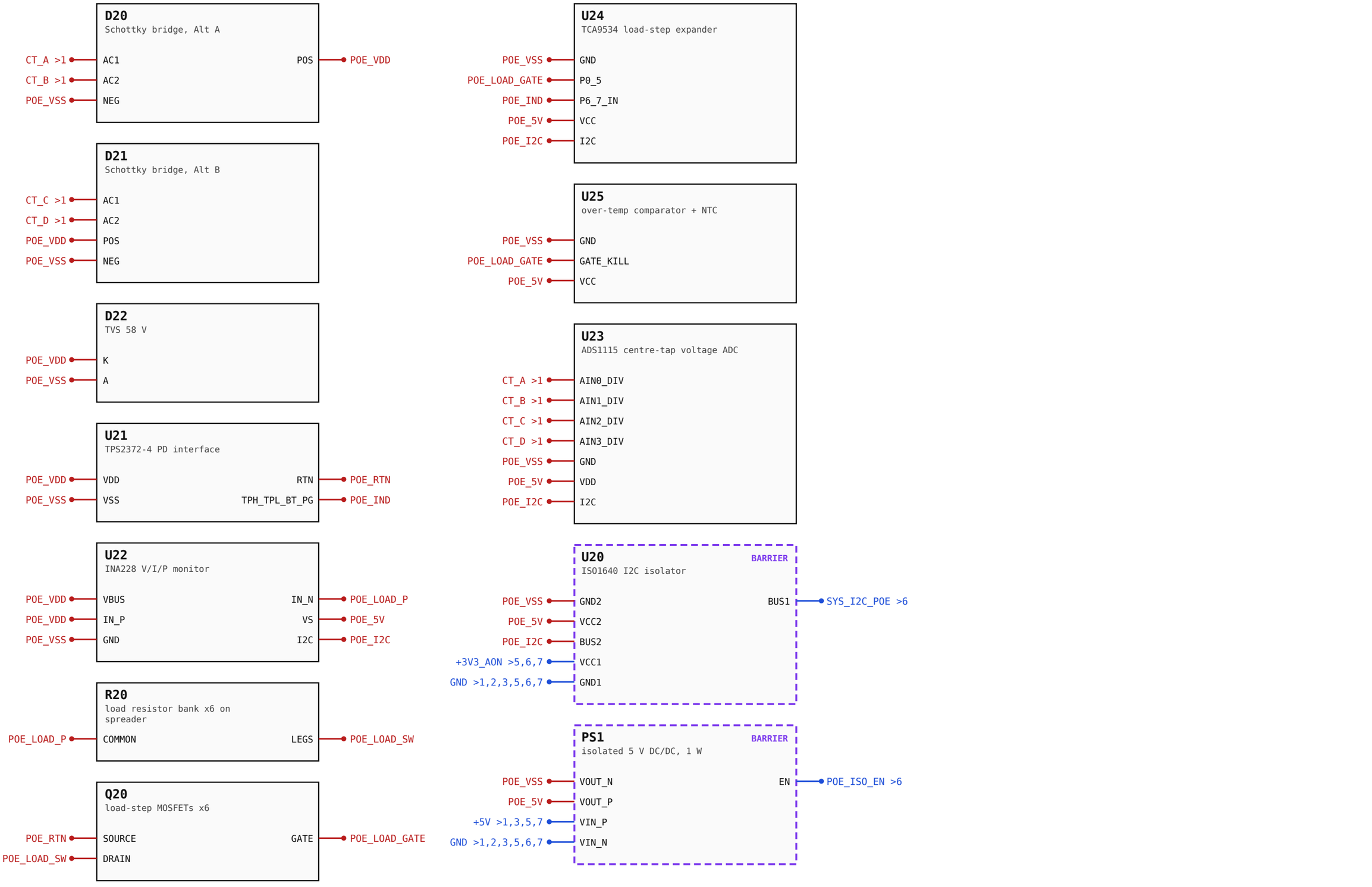
NetScope X1 main board - block schematic, rev A draft

Wiremap front end

Pre-hardware. Not a KiCad schematic. Not ERC-checked. No board exists.

SYS system side LINE cable conductors POE network-powered

Block level: function-named pins, not package pins



NetScope X1 main board - block schematic, rev A draft

PoE measurement island (isolated)

Pre-hardware. Not a KiCad schematic. Not ERC-checked. No board exists.

SYS system side · LINE cable conductors · POE network-powered

Block level: function-named pins, not package pins

M1
Raspberry Pi CM5 wireless

USB3_1 >2	USB3_0	UART	COPROC_UART >6
B_MDI >2	ETH_MDI	GPIO_COPROC_BOOT_RST	COPROC_BOOT >6
B_LEDS >2	ETH_LEDS	UART_DEBUG	SOM_UART_DBG
+5V >1,3,4,7	5V	NRPIBOOT	SOM_NRPIBOOT
+3V3_AON >4,6,7	VBAT_RTC	PMIC_EN_RUN	SOM_RUN >6
GND >1,2,3,4,6,7	GND	DSI1	DSI
		I2C_DISPLAY	TOUCH_I2C
		USB2_OTG	USBC_D >7
		USB3_1_D	USB3_2_HS
		PCIE	PCIE
		I2C_SFP	SFP_I2C

J1
DSI FFC to 5in touch display

DSI	DSI
TOUCH_I2C	I2C
+5V >1,3,4,7	5V
GND >1,2,3,4,6,7	GND

U70
AQC100 10G MAC, PCIe to SFP+

+5V >1,3,4,7	VIN_VIA_REG	XFI_TX_RX	SFP_XFI
GND >1,2,3,4,6,7	GND	SFP_LOS	SFP_STATUS >6
PCIE	PCIE	PWR_EN	SFP_PWR_EN >6

J32
SFP+ cage and 20-pin connector

+5V >1,3,4,7	VCC_VIA_SWITCH	TX_DISABLE	SFP_TX_DISABLE >6
GND >1,2,3,4,6,7	GND_AND_CAGE		
SFP_XFI	TD_RD		
SFP_I2C	MOD_DEF_SDA_SCL		
SFP_STATUS >6	LOS_PRESENT_TXFAULT		
SFP_PWR_EN >6	VCC_SWITCH_EN		

U60
USB2244 microSD reader (DNP option)

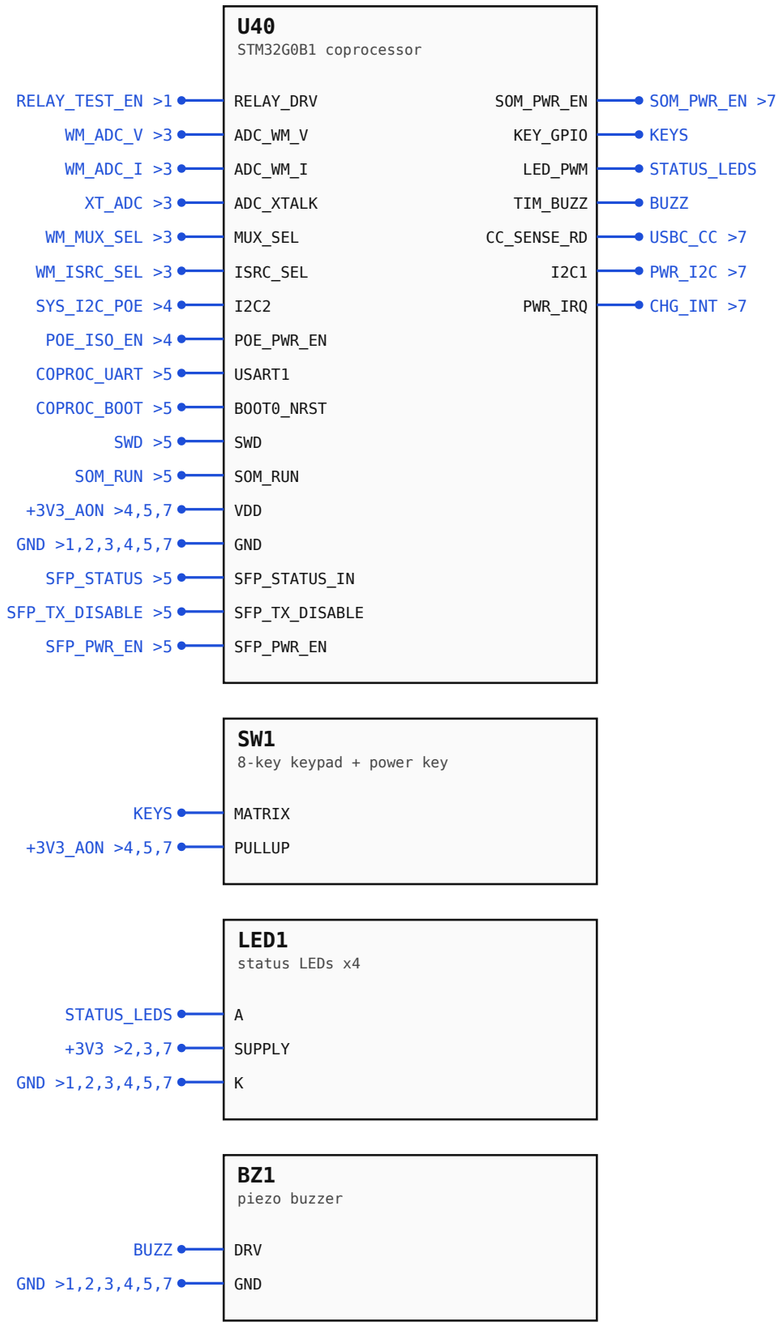
+5V >1,3,4,7	VBUS_IN	SD	SD_BUS
GND >1,2,3,4,6,7	GND		
USB3_2_HS	USB		

J30
microSD socket (DNP option)

GND >1,2,3,4,6,7	GND
SD_BUS	SD

J40
debug: UART + SWD + boot header

COPROC_UART >6	UART_TAP	SWD	SWD >6
SOM_UART_DBG	CONSOLE		
SOM_NRPIBOOT	RPIBOOT_JUMPER		
GND >1,2,3,4,6,7	GND		



NetScope X1 main board - block schematic, rev A draft

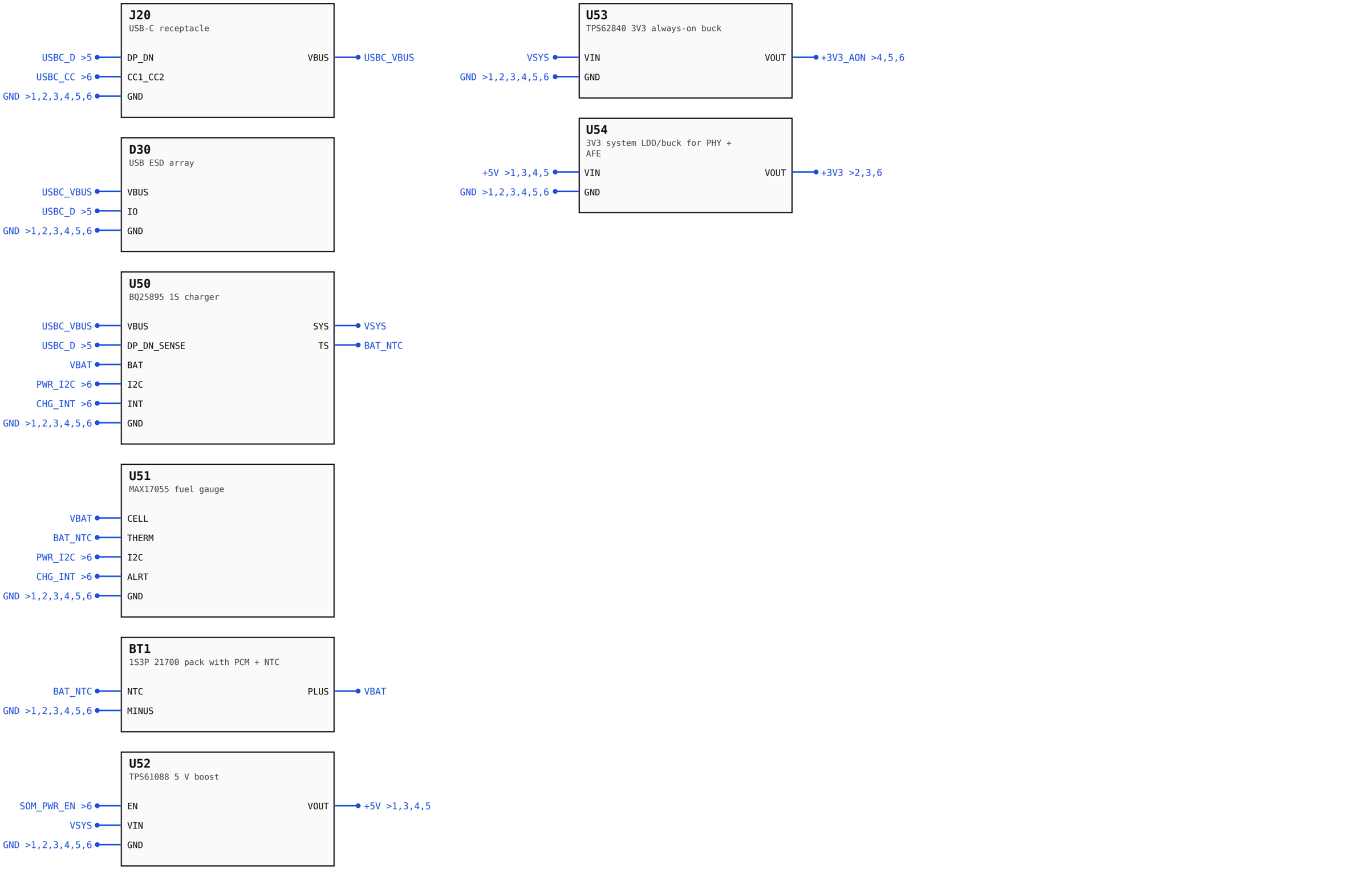
Coprocessor, keys, indicators

Pre-hardware. Not a KiCad schematic. Not ERC-checked. No board exists.

SYS system side LINE cable conductors POE network-powered

Sheet 6 of 10

Block level: function-named pins, not package pins



NetScope X1 main board - block schematic, rev A draft

Power: USB-C, charger, battery, rails

Pre-hardware. Not a KiCad schematic. Not ERC-checked. No board exists.

SYS system side · LINE cable conductors · POE network-powered

Block level: function-named pins, not package pins

Net	Domain	Nodes
A_L1	LINE	J10.P1, K1.COM_A
A_L2	LINE	J10.P2, K1.COM_B
A_L3	LINE	J10.P3, K2.COM_A
A_L6	LINE	J10.P6, K2.COM_B
A_L4	LINE	J10.P4, K3.COM_A
A_L5	LINE	J10.P5, K3.COM_B
A_L7	LINE	J10.P7, K4.COM_A
A_L8	LINE	J10.P8, K4.COM_B
A_M1	LINE	K1.NC_A, T1.LINE_A_P
A_M2	LINE	K1.NC_B, T1.LINE_A_N
A_M3	LINE	K2.NC_A, T1.LINE_B_P
A_M6	LINE	K2.NC_B, T1.LINE_B_N
A_M4	LINE	K3.NC_A, T1.LINE_C_P
A_M5	LINE	K3.NC_B, T1.LINE_C_N
A_M7	LINE	K4.NC_A, T1.LINE_D_P
A_M8	LINE	K4.NC_B, T1.LINE_D_N
A_T1	LINE	K1.NO_A, PROT1.IN1
A_T2	LINE	K1.NO_B, PROT1.IN2
A_T3	LINE	K2.NO_A, PROT1.IN3
A_T6	LINE	K2.NO_B, PROT1.IN6
A_T4	LINE	K3.NO_A, PROT1.IN4
A_T5	LINE	K3.NO_B, PROT1.IN5
A_T7	LINE	K4.NO_A, PROT1.IN7
A_T8	LINE	K4.NO_B, PROT1.IN8
RELAY_TEST_EN	SYS	U40.RELAY_DRV, K1.COIL, K2.COIL, K3.COIL, K4.COIL
WM1	SYS	PROT1.OUT1, U30.S1, U31.S1
WM2	SYS	PROT1.OUT2, U30.S2, U31.S2
WM3	SYS	PROT1.OUT3, U30.S3, U31.S3
WM4	SYS	PROT1.OUT4, U30.S4, U31.S4
WM5	SYS	PROT1.OUT5, U30.S5, U31.S5
WM6	SYS	PROT1.OUT6, U30.S6, U31.S6
WM7	SYS	PROT1.OUT7, U30.S7, U31.S7
WM8	SYS	PROT1.OUT8, U30.S8, U31.S8
WM_DRIVE	SYS	U30.D, U32.SRC_OUT, U33.TONE_OUT
WM_RETURN	SYS	U31.D, U32.SENSE_IN, U33.DET_IN
WM_ADC_V	SYS	U32.V_OUT, U40.ADC_WM_V
WM_ADC_I	SYS	U32.I_OUT, U40.ADC_WM_I
XT_ADC	SYS	U33.DET_OUT, U40.ADC_XTALK
WM_MUX_SEL	SYS	U40.MUX_SEL, U30.A, U31.A
WM_ISRC_SEL	SYS	U40.ISRC_SEL, U32.RANGE
A_MDI_A	SYS	T1.PHY_A, U10.TXRX_A, D10.IO_A
A_MDI_B	SYS	T1.PHY_B, U10.TXRX_B, D10.IO_B
A_MDI_C	SYS	T1.PHY_C, U10.TXRX_C, D10.IO_C
A_MDI_D	SYS	T1.PHY_D, U10.TXRX_D, D10.IO_D

Net	Domain	Nodes
A_RGMII	SYS	U10.RGMII, U11.RGMII
A_MDIO	SYS	U10.MDIO, U11.MDIO
A_PHY_RST_N	SYS	U11.PHY_RESET_N, U10.RESET_N
USB3_1	SYS	M1.USB3_0, U11.USB
CT_A	POE	T1.CT_A, D20.AC1, U23.AIN0_DIV
CT_B	POE	T1.CT_B, D20.AC2, U23.AIN1_DIV
CT_C	POE	T1.CT_C, D21.AC1, U23.AIN2_DIV
CT_D	POE	T1.CT_D, D21.AC2, U23.AIN3_DIV
POE_VDD	POE	D20.POS, D21.POS, D22.K, U21.VDD, U22.VBUS, U22.IN_P
POE_VSS	POE	D20.NEG, D21.NEG, D22.A, U21.VSS, U23.GND, U22.GND, U24.GND, U20.GND2, PS1.VOUT_N, U25.GND
POE_LOAD_P	POE	U22.IN_N, R20.COMMON
POE_RTN	POE	U21.RTN, Q20.SOURCE
POE_LOAD_SW	POE	R20.LEGS, Q20.DRAIN
POE_LOAD_GATE	POE	U24.P0_5, Q20.GATE, U25.GATE_KILL
POE_IND	POE	U21.TPH_TPL_BT_PG, U24.P6_7_IN
POE_5V	POE	PS1.VOUT_P, U22.VS, U23.VDD, U24.VCC, U20.VCC2, U25.VCC
POE_I2C	POE	U20.BUS2, U22.I2C, U23.I2C, U24.I2C
SYS_I2C_POE	SYS	U40.I2C2, U20.BUS1
POE_ISO_EN	SYS	U40.POE_PWR_EN, PS1.EN
B_MDI	SYS	M1.ETH_MDI, J11.PHY_SIDE
B_LEDS	SYS	M1.ETH_LEDS, J11.LEDS
COPROC_UART	SYS	M1.UART, U40.USART1, J40.UART_TAP
COPROC_BOOT	SYS	M1.GPIO_COPROC_BOOT_RST, U40.BOOT0_NRST
SWD	SYS	U40.SWD, J40.SWD
SOM_UART_DBG	SYS	M1.UART_DEBUG, J40.CONSOLE
SOM_NRPIBOOT	SYS	M1.NRPIBOOT, J40.RPIBOOT_JUMPER
SOM_PWR_EN	SYS	U40.SOM_PWR_EN, U52.EN
SOM_RUN	SYS	U40.SOM_RUN, M1.PMIC_EN_RUN
KEYS	SYS	SW1.MATRIX, U40.KEY_GPIO
STATUS_LEDS	SYS	U40.LED_PWM, LED1.A
BUZZ	SYS	U40.TIM_BUZZ, BZ1.DRV
DSI	SYS	M1.DSI1, J1.DSI
TOUCH_I2C	SYS	M1.I2C_DISPLAY, J1.I2C
USBC_VBUS	SYS	J20.VBUS, U50.VBUS, D30.VBUS
USBC_D	SYS	J20.DP_DN, D30.IO, U50.DP_DN_SENSE, M1.USB2_OTG
USBC_CC	SYS	J20.CC1_CC2, U40.CC_SENSE_RD
VBAT	SYS	BT1.PLUS, U51.CELL, U50.BAT
VSYS	SYS	U50.SYS, U52.VIN, U53.VIN
BAT_NTC	SYS	BT1.NTC, U50.TS, U51.THERM
+5V	SYS	U52.VOUT, M1.5V, J1.5V, U54.VIN, U55.VIN, PS1.VIN_P, K1.COIL_SUPPLY, K2.COIL_SUPPLY, K3.COIL_SUPPLY, K4.COIL_SUPPLY, U60.VBUS_IN, U70.VIN_VIA_REG, J32.VCC_VIA_SWITCH
+3V3_AON	SYS	U53.VOUT, U40.VDD, M1.VBAT_RTC, U20.VCC1, SW1.PULLUP
+3V3	SYS	U54.VOUT, U10.VDD, U11.VDD, U32.VDD, U33.VDD, LED1.SUPPLY
+15V	SYS	U55.VPOS, U30.VDD, U31.VDD
-15V	SYS	U55.VNEG, U30.VSS, U31.VSS

Net	Domain	Nodes
PWR_I2C	SYS	U40.I2C1, U50.I2C, U51.I2C
CHG_INT	SYS	U50.INT, U51.ALRT, U40.PWR_IRQ
GND	SYS	M1.GND, U10.GND, U11.GND, U30.GND, U31.GND, U32.GND, U33.GND, U40.GND, U50.GND, U51.GND, U52.GND, U53.GND, U54.GND, U55.GND, BT1.MINUS, J20.GND, D10.GND, D30.GND, PR...
CHASSIS	LINE	J10.SHIELD, J11.SHIELD, T1.BOB_SMITH_HV_CAP
USB3_2_HS	SYS	M1.USB3_1_D, U60.USB
SD_BUS	SYS	U60.SD, J30.SD
PCIE	SYS	M1.PCIE, U70.PCIE
SFP_XFI	SYS	U70.XFI_TX_RX, J32.TD_RD
SFP_I2C	SYS	M1.I2C_SFP, J32.MOD_DEF_SDA_SCL
SFP_STATUS	SYS	J32.LOS_PRESENT_TXFAULT, U40.SFP_STATUS_IN, U70.SFP_LOS
SFP_TX_DISABLE	SYS	U40.SFP_TX_DISABLE, J32.TX_DISABLE
SFP_PWR_EN	SYS	U40.SFP_PWR_EN, U70.PWR_EN, J32.VCC_SWITCH_EN